Paper Code:

Measurement-Oriented Microscopy Reconstruction for Cross-Magnification Micro-Displacement Inspection

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Introduction

Microscopy reconstruction supports optical inspection when clearer boundaries also produce more stable position measurements. Objective changes alter object-space sampling, optical blur and the pixel phase of motion. The variation of measured single steps provides a direct test of reconstruction for displacement inspection.

MOSR reconstructs each monochrome image on its native pixel grid. One fixed set of learned weights serves 20× and 50× objectives. This study examines each objective at a common nominal step of 1 µm, with Raw, fixed Wiener and MOSR sharing the same image pairs and circle-center calculation.

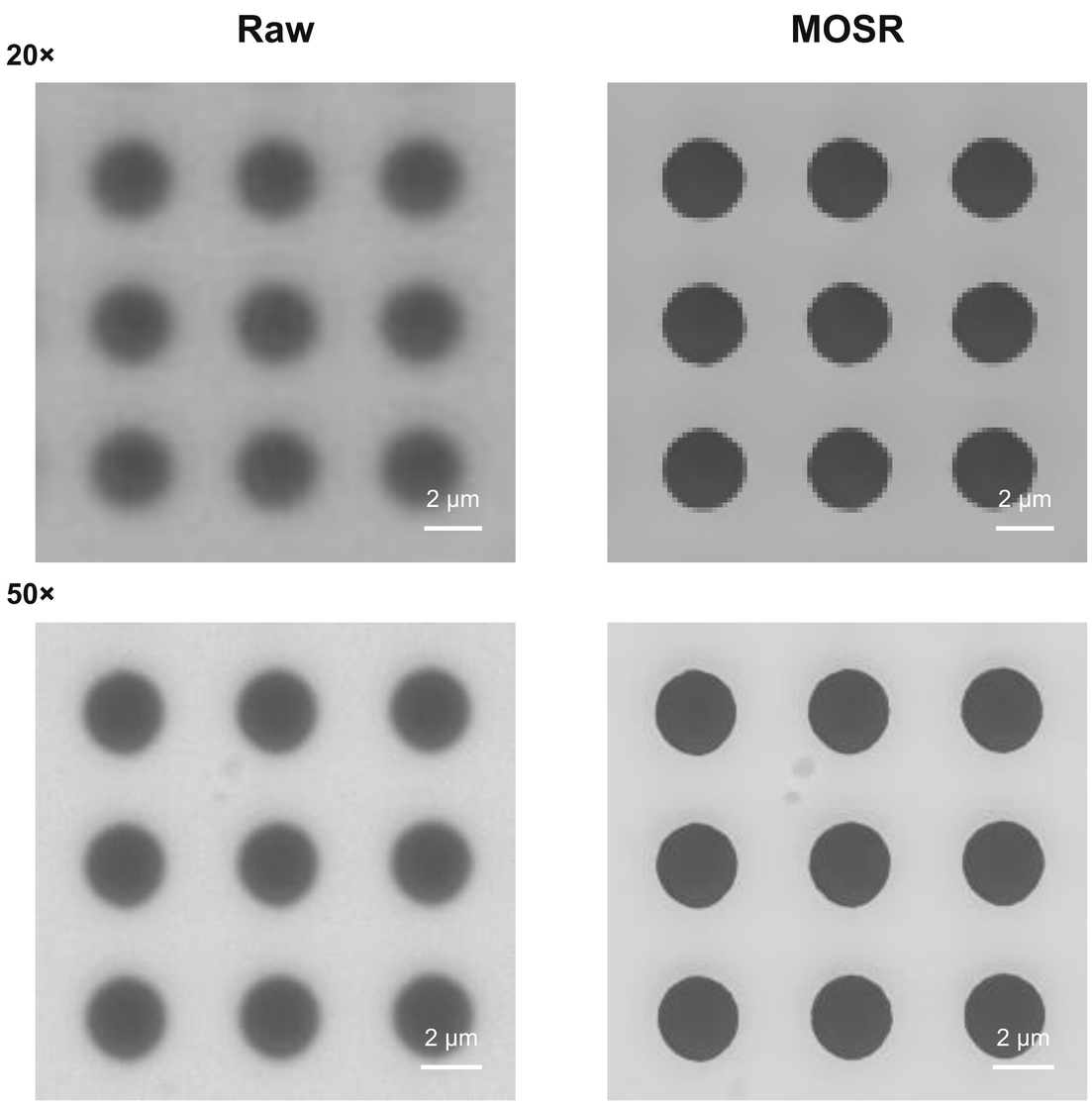


Fig. 1. Raw and MOSR images from the 20× and 50×, 1 µm datasets. Each comparison uses the same field and grayscale range. Scale bars: 2 µm.

Methods

Reconstruction. MOSR combines U-Net context [1], anti-aliasing [2] and bounded PSF guidance. The optical inputs comprise 25 measured kernels at 20× and one calibrated kernel at 50×. Fixed Wiener uses a regularization value of 0.0565.

Localization. Coarse contours [3] initialize maximum-gradient boundary points. Geometric least-squares circle fitting [4] uses 64 points at 20× and 96 at 50×. Gradient peaks provide subpixel boundary positions.

Displacement. The median of nine pillar-center changes defines each step along the acquisition motion direction. The 20× dataset contains 30 complete scans of 30 steps. The 50× dataset contains 30 locked adjacent-frame pairs.

TABLE I. SEPARATE ACQUISITION CONDITIONS

Objective	Step (µm)	Measured steps	µm/pixel
20×	1.0	30 scans × 30	0.1725
50×	1.0	30 pairs	0.0690

Each scan or locked set yields the sample SD of 30 steps (denominator 29). At 20×, all 30 scan SDs are shown; the within-scan summary is the square root of their mean squared SD. The 50× result is reported separately. Both use image-measured displacement.

Results

At a common 1 µm step, MOSR lowers displacement variation at both objectives. Results are calculated within each acquisition condition.

20×, 1 µm: 30 repeated scans

52.3%

Reduction in pooled within-scan Step SD

10.200 nm Raw
4.865 nm MOSR

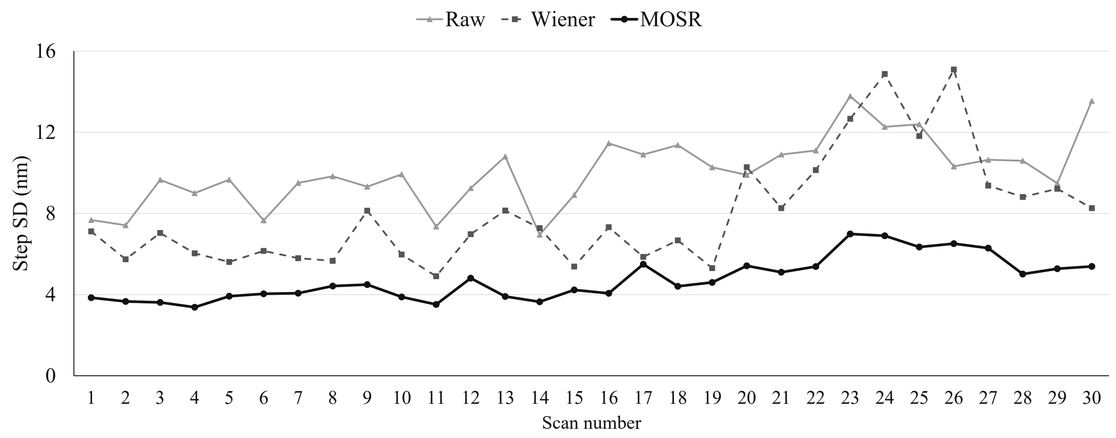


Fig. 2. Each point is the SD of 30 measured steps in one 20× scan. The 30 scans retain their individual results for all three methods.

MOSR reduces Step SD in all 30 scans, with per-scan reductions of 36.9–64.6% relative to Raw. The within-scan summary is 8.431 nm for fixed Wiener and 4.865 nm for MOSR.

50×, 1 µm: 30 locked pairs

23.5%

Reduction in Step SD

5.938 nm Raw
4.542 nm MOSR

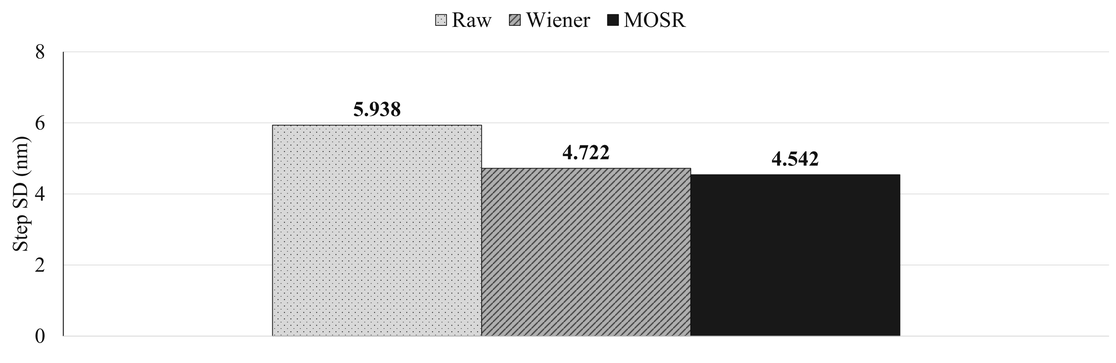


Fig. 3. SD of the 30 locked 1 µm displacements at 50×. The same calibrated optical input supports Wiener and MOSR; the circle-center calculation is shared by all methods.

Conclusions

At 20×, MOSR lowers pooled within-scan step variation by 52.3% and improves all 30 repeated scans. At 50×, Step SD falls by 23.5% for the fixed 1 µm condition.

One reconstruction model supports both objectives with calibrated optical inputs. Clearer boundaries and lower displacement variation support cross-magnification optical inspection.

Bibliography

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